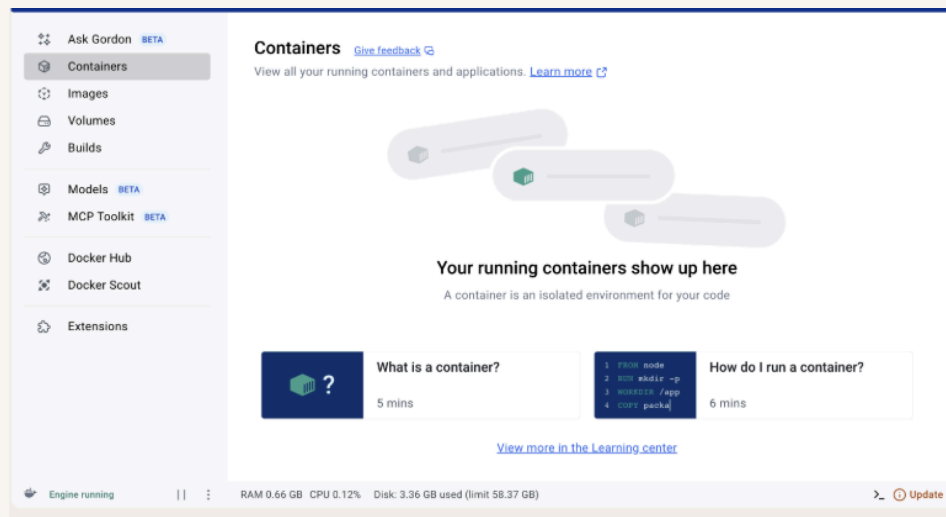




Deploy an App with Docker



Michael K.





Introducing Today's Project!

What is Docker?

Docker is a platform for building, packaging, and running applications in containers. A container is a lightweight, portable, self-contained environment that includes: The application code, Dependencies (libraries, runtime, etc.) and Configuration files. It is useful because it comes with everything it needs, and you can move it anywhere without worrying about the environment at the destination. I used Docker in this project to create an image of my web App, build it, and test it.

One thing I didn't expect...

One thing I didn't expect in this project was its complexity given the fact that i did it in a private EC2 Instance server.

This project took me...

This project took me approximately 14hours. The most challenging part was to build and test the App in the Private EC2 Instance server. It was most rewarding to succeed at the end.



Understanding Containers and Docker

Containers

Containers packages Applications and their dependencies in one File (Container Images), and also provide an environment to deploy and run Applications.

A container image is a blueprint or template for making containers. It tells Docker exactly what to put inside each container - things like app's code, the libraries it needs, and any other required files.

Docker

Docker helps me create, manage, and deploy containers efficiently. It is like a tool that helps you create the cargo and load them onto ships.

The Docker daemon is a background process that manages the Docker containers on the computer. It takes commands from the Docker client (i.e. commands from the terminal, or clicks from the Docker Desktop) and does the heavy lifting of building, running, and distributing your containers.



Running an Nginx Image

Nginx (pronounced “engine-x”) is a web server, which is a program i use to run websites and web apps. If i want people to visit my site, i'm going to need a web server to deliver my website's files to their browsers! Engineers pick Nginx when they need to manage lots of visitors at once, making sure a site doesn't slow down when traffic spikes.

The command I ran to start a new container was << docker run >>

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Creating a Custom Image

The Dockerfile is a document with all the instructions for building a Docker image. Docker would read a Dockerfile to understand how to set up the application's environment and which software packages it should install.

My Dockerfile tells Docker three things : FROM nginx:latest, COPY index.html /usr/share/nginx/html/, EXPOSE 80

The command I used to build a custom image with my Dockerfile was << docker build -t my-web-app . >> . The '.' at the end of the command tells Docker to find the Dockerfile in the current directory i.e. the Compute folder.



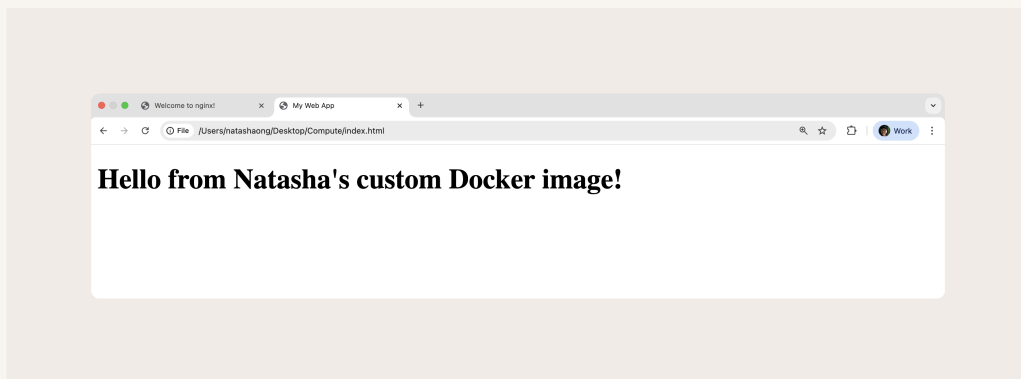
```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/
EXPOSE 80
```



Running My Custom Image

There was an error when I ran my custom image because there's already a container using port 80, so the new container i'm creating can't access it. I resolved this by stopping that already running container.

In this example, the container image is `<<nginx>>` and the container is the one with the ID.

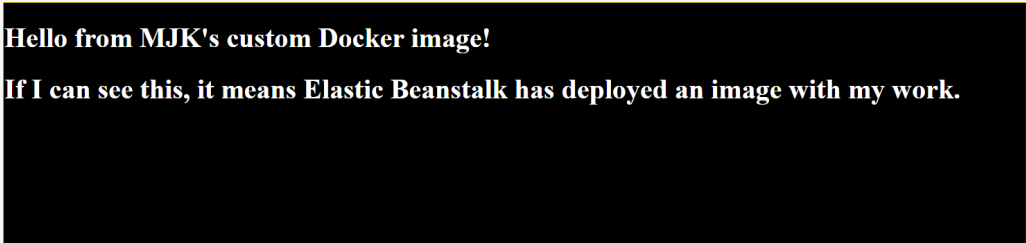




Elastic Beanstalk

Elastic Beanstalk is a service that makes it easy to deploy cloud applications without worrying about the underlying infrastructure. Simply upload the code and Elastic Beanstalk handles everything needed to get it running, like setting up servers and managing scaling.

Deploying my custom image with Elastic Beanstalk took me 6mins



```

Hello from MJK's custom Docker image!
If I can see this, it means Elastic Beanstalk has deployed an image with my work.
```



Deploying App Updates

To learn how to deploy app updates with Elastic Beanstalk, I updated my app by customizing my web Page. I verified those changes by opening my local index.html file in my Instance.

My app updates didn't show up in my live environment immediately because the change was only locally to be seen. To deploy my changes, I only had to upload the new version of the file in S3 and after rebuild my App in Elastic Beanstalk.

Hello from MJK's custom Docker image!

If I can see this, it means Elastic Beanstalk has deployed an image with my work.

And here's a special image chosen by me:

